

Tail-projection certificates for cyclotomic minimum-degree polynomials beyond the reciprocal condition

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Abstract

Let $\Phi_n(x)$ be the n -th cyclotomic polynomial, and let p be the smallest prime divisor of n . Steinberger formulated as Conjecture 1 the assertion that the unique monic polynomial of lowest degree among the nonzero polynomials in $\mathbb{R}_{\geq 0}[x]$ divisible by Φ_n is

$$R_{n,p}(x) = 1 + x^{n/p} + \cdots + x^{(p-1)n/p}.$$

He explicitly expressed skepticism about this assertion and anticipated counterexamples. For an odd squarefree integer $n = pq_1 \cdots q_k$, where $p < q_1 < \cdots < q_k$ and $k \geq 3$, we give an explicit sufficient condition for this conclusion. Put $P = q_1 \cdots q_k$. For every $d \mid P$, let $\rho_d \in \{1, \dots, p-1\}$ be congruent to P/d modulo p , and write $\sigma_d = (-1)^{k-\omega(d)}$. We prove that

$$\sum_{\substack{d \mid P, d < P \\ \sigma_d = -1}} d\rho_d + p \sum_{\substack{d \mid P, d < P \\ \sigma_d = +1}} d + \frac{p}{q_1 - 1} \sum_{\substack{d \mid P \\ d < P}} d < P$$

implies that the minimum degree is $(p-1)P$, with equality only for positive scalar multiples of $R_{n,p}$. The proof constructs an integer separation certificate by applying a zero-marginal CRT tensor projection to the final interval of $P-1$ exponents and then cancelling its values on the regular polygon by an explicit anchor correction.

For every fixed $k \geq 3$, the new condition holds for all sufficiently large prime tuples

$$p < q_1 < \cdots < q_k \leq (1 + \delta)p, \quad 0 < \delta < 1/k.$$

These tuples fail Steinberger's reciprocal condition $2/p > \sum_i 1/q_i$. Consequently the theorem supplies, for each fixed number of at least four distinct prime factors, infinitely many cases outside the range of the previously known reciprocal sufficient condition.

1 Introduction

For a positive integer n , consider

$$\mu(n) = \min\{\deg f : 0 \neq f \in \mathbb{R}_{\geq 0}[x], \Phi_n(x) \mid f(x)\}. \quad (1.1)$$

If p is the smallest prime divisor of n , then

$$R_{n,p}(x) = 1 + x^{n/p} + \cdots + x^{(p-1)n/p} \quad (1.2)$$

is a nonnegative multiple of Φ_n , and hence

$$\mu(n) \leq (p-1)\frac{n}{p}. \quad (1.3)$$

Steinberger [1] named equality in (1.3), together with uniqueness of the monic minimizer, Conjecture 1, while explicitly stating that he did not believe it in full generality and expected counterexamples. For a squarefree factorization

$$n = pq_1 \cdots q_k, \quad p < q_1 < \cdots < q_k, \quad (1.4)$$

his main theorem proves the conjecture under

$$\frac{2}{p} > \sum_{i=1}^k \frac{1}{q_i}. \quad (1.5)$$

The 2012 paper left the general problem open. To the best of our knowledge, the literature search described in Section 8 found no subsequent complete resolution. A 2025 treatment of the corresponding integer-tiling diameter problem still cites (1.5) as the relevant deeper sufficient theorem and emphasizes the additional structural difficulty once three or more non-minimal prime factors are present [2].

The present paper proves a different sufficient condition. Its origin is a family of exact finite certificates developed after the first two parameter values not covered by (1.5) were settled computationally [3]. The result below no longer requires a linear program or a stored high-dimensional certificate: a single interval, a tensor projection, and a closed divisor–residue estimate provide a certificate for an infinite family.

Theorem 1.1. *Let $n = pq_1 \cdots q_k$ be the complete factorization of n into distinct odd primes, with*

$$p < q_1 < \cdots < q_k, \quad k \geq 3,$$

and put $P = q_1 \cdots q_k$. For every divisor $d \mid P$, define

$$\rho_d \in \{1, \dots, p-1\}, \quad \rho_d \equiv P/d \pmod{p}, \quad (1.6)$$

and

$$\sigma_d = (-1)^{k-\omega(d)}, \quad (1.7)$$

where $\omega(d)$ is the number of distinct prime factors of d . Set

$$\mathcal{B}_k = \sum_{\substack{d \mid P, \, d < P \\ \sigma_d = -1}} d \rho_d + p \sum_{\substack{d \mid P, \, d < P \\ \sigma_d = +1}} d + \frac{p}{q_1 - 1} \sum_{\substack{d \mid P \\ d < P}} d. \quad (1.8)$$

If

$$\mathcal{B}_k < P, \tag{1.9}$$

then every nonzero $f \in \mathbb{R}_{\geq 0}[x]$ divisible by Φ_n satisfies

$$\deg f \geq (p-1)P. \tag{1.10}$$

Equality holds if and only if $f = cR_{n,p}$ for some $c > 0$. In particular, $\mu(n) = (p-1)P$, and $R_{n,p}$ is the unique monic minimizer.

Theorem 1.2. Fix $k \geq 3$ and $0 < \delta < 1/k$. Among the prime tuples satisfying

$$p < q_1 < \cdots < q_k \leq (1 + \delta)p, \tag{1.11}$$

all sufficiently large tuples satisfy (1.9), while all such tuples fail (1.5). There are infinitely many such tuples.

In particular, if p_m denotes the m -th prime, then

$$(p, q_1, \dots, q_k) = (p_m, p_{m+1}, \dots, p_{m+k}) \tag{1.12}$$

satisfies (1.9) and fails (1.5) for all sufficiently large m .

Section 2 recalls the separation criterion and boundary rigidity. Sections 3–5 construct and estimate the certificate. Section 6 proves the infinite family. Section 7 records a four-prime corollary and exact calibration, and Section 8 explains the scope relative to earlier work.

2 Boundary separation and rigidity

Let n be squarefree, let p be its smallest prime divisor, and put

$$P = n/p, \quad H = \{0, P, \dots, (p-1)P\}, \tag{2.1}$$

and

$$F^+ = \{0, \dots, (p-1)P - 1\} \setminus H. \tag{2.2}$$

Let $V_n \subseteq \mathbb{R}^n$ be the coefficient-vector space of polynomials of degree less than n divisible by Φ_n . We use the sign-reversed form of Steinberger's separation criterion [1, Lemma 1].

Proposition 2.1. *Suppose that $w \in V_n^\perp$ satisfies*

$$w_j = 0 \quad (j \in H), \quad w_j < 0 \quad (j \in F^+). \quad (2.3)$$

Then every nonzero $f \in \mathbb{R}_{\geq 0}[x]$ divisible by Φ_n and of degree at most $(p-1)P$ is supported on H .

Proof. Pad the coefficient vector a of f with zeros to length n . Then $a \in V_n$, so

$$0 = \langle w, a \rangle = \sum_{j=0}^{(p-1)P} w_j a_j.$$

All summands are nonpositive. A positive coefficient at any index in F^+ would make the sum strictly negative. Hence $a_j = 0$ for every $j \in F^+$. $\square$

Lemma 2.2. *If $0 \neq f \in \mathbb{R}_{\geq 0}[x]$, $\Phi_n \mid f$, $\deg f \leq (p-1)P$, and $\text{supp}(f) \subseteq H$, then*

$$f(x) = cR_{n,p}(x) \quad (2.4)$$

for some $c > 0$.

Proof. Write $f(x) = g(x^P)$, where $\deg g \leq p-1$. Let ζ_n be a primitive n -th root. For every $a \in (\mathbb{Z}/n\mathbb{Z})^*$,

$$0 = f(\zeta_n^a) = g((\zeta_n^P)^a).$$

The reduction map from $(\mathbb{Z}/n\mathbb{Z})^*$ onto $(\mathbb{Z}/p\mathbb{Z})^*$ is surjective by the Chinese remainder theorem. Thus g vanishes at every primitive p -th root, so $\Phi_p \mid g$ in $\mathbb{R}[x]$. The degree bound forces $g = c\Phi_p$, and nonzero nonnegative coefficients imply $c > 0$. $\square$

3 CRT projection and anchor correction

Assume the notation of Theorem 1.1. Identify $\mathbb{Z}_n$ with

$$G = \mathbb{Z}_p \times \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_k} \quad (3.1)$$

through the exponent bijection

$$(a, b_1, \dots, b_k) \mapsto aP + \sum_{i=1}^k b_i \frac{n}{q_i} \pmod{n}. \quad (3.2)$$

Under this identification, H is $\{(a, 0, \dots, 0) : a \in \mathbb{Z}_p\}$. For a coordinate of length t , let I_t and J_t denote the identity and all-ones matrices. Define

$$D = \bigotimes_{t \in \{p, q_1, \dots, q_k\}} (tI_t - J_t). \quad (3.3)$$

Each factor has image $\mathbf{1}_t^\perp$. The standard CRT fiber description of V_n gives

$$\text{im } D = \mathbf{1}_p^\perp \otimes \mathbf{1}_{q_1}^\perp \otimes \cdots \otimes \mathbf{1}_{q_k}^\perp = V_n^\perp. \quad (3.4)$$

For completeness, we justify the final equality. Let W be the span of all coordinate-fiber indicators in G . A fiber in the coordinate of a prime $r \mid n$ is, up to a cyclic translation, the coefficient vector of

$$1 + x^{n/r} + \cdots + x^{(r-1)n/r} = \frac{x^n - 1}{x^{n/r} - 1},$$

so it belongs to V_n . Hence $W \subseteq V_n$. An array is orthogonal to every coordinate fiber exactly when every coordinate-fiber sum is zero; therefore

$$W^\perp = \mathbf{1}_p^\perp \otimes \mathbf{1}_{q_1}^\perp \otimes \cdots \otimes \mathbf{1}_{q_k}^\perp.$$

The right-hand side has dimension $(p-1) \prod_i (q_i - 1) = \varphi(n)$, so $\dim W = n - \varphi(n)$. On the other hand, multiplication by Φ_n identifies the polynomials of degree less than $n - \varphi(n)$ with V_n , and hence $\dim V_n = n - \varphi(n)$. Thus $W = V_n$ and the final equality in (3.4) follows. This is also the fiber description in [1, Section 2, pp. 5–7].

Let

$$\mathcal{I} = \{n - P + 1, \dots, n - 1\}, \quad u = D\mathbf{1}_{\mathcal{I}}, \quad C = \prod_{i=1}^k (q_i - 1). \quad (3.5)$$

For $a \in \mathbb{Z}_p$, put $u_a = u(a, 0, \dots, 0)$. Because the p -coordinate factor is $pI_p - J_p$,

$$\sum_{a \in \mathbb{Z}_p} u_a = 0. \quad (3.6)$$

Define

$$Y = pC\mathbf{1}_{\mathcal{I}} - \sum_{a \in \mathbb{Z}_p} u_a \mathbf{1}_{\{(a, 0, \dots, 0)\}}, \quad w = DY. \quad (3.7)$$

Then Y, w are integer arrays and $w \in V_n^\perp$.

Lemma 3.1. *Define*

$$\varepsilon(x_{q_1}, \dots, x_{q_k}) = \prod_{i=1}^k \frac{q_i \mathbf{1}_{x_{q_i}=0} - 1}{q_i - 1}. \quad (3.8)$$

Then

$$\frac{w(x)}{pC} = u(x) - u_{x_p} \varepsilon(x_{q_1}, \dots, x_{q_k}). \quad (3.9)$$

Consequently $w = 0$ on H . Away from H ,

$$|\varepsilon| \leq \frac{1}{q_1 - 1}. \quad (3.10)$$

Proof. For an anchor $e_a = (a, 0, \dots, 0)$,

$$D\mathbf{1}_{\{e_a\}}(x) = (p\mathbf{1}_{x_p=a} - 1) \prod_{i=1}^k (q_i \mathbf{1}_{x_{q_i}=0} - 1). \quad (3.11)$$

Use (3.6), multiply by u_a , and sum over a to obtain (3.9). At an anchor, all factors in (3.8) equal 1 and $u(x) = u_{x_p}$. At a nonanchor, at least one final coordinate is nonzero, producing a factor of magnitude $1/(q_i - 1)$; all other factors have magnitude at most 1. This proves the two assertions. $\square$

4 Exact projection of the final interval

For $m \mid n$ and $j \in \mathbb{Z}_n$, let

$$N_m(j) = \#\{\ell \in \mathcal{I} : \ell \equiv j \pmod{m}\}. \quad (4.1)$$

Expanding (3.3) and pairing the terms indexed by d and pd gives

$$u(j) = \sum_{d \mid P} (-1)^{k-\omega(d)} d e_d(j), \quad e_d(j) = pN_{pd}(j) - N_d(j). \quad (4.2)$$

Lemma 4.1. *For every $d \mid P$,*

$$N_d(j) = \frac{P}{d} - \mathbf{1}_{d \mid j}, \quad (4.3)$$

and

$$N_{pd}(j) = \left\lfloor \frac{P - 1 + (j \bmod pd)}{pd} \right\rfloor. \quad (4.4)$$

If $P/d = ap + \rho_d$, then

$$e_d(j) = p\theta_d(j) - \rho_d + \mathbf{1}_{d \mid j} \quad (4.5)$$

for some $\theta_d(j) \in \{0, 1\}$. In particular,

$$-\rho_d \leq e_d(j) \leq p. \quad (4.6)$$

Proof. Write $\ell = n - r$, where $1 \leq r \leq P - 1$. Every residue modulo d occurs P/d times in $\{0, \dots, P - 1\}$; removing $r = 0$ gives (4.3). Counting the positive integers $r \leq P - 1$ congruent to $-j$ modulo pd gives (4.4). The integer in (4.4) is either a or $a + 1$. Substitution into e_d proves (4.5), and $1 \leq \rho_d \leq p - 1$ yields (4.6). $\square$

Lemma 4.2. *For every $j \in F^+$,*

$$u(j) \leq -P + \sum_{\substack{d \mid P, d < P \\ \sigma_d = -1}} d\rho_d + p \sum_{\substack{d \mid P, d < P \\ \sigma_d = +1}} d. \quad (4.7)$$

For every anchor $h \in H$,

$$|u(h)| \leq p \sum_{\substack{d \mid P \\ d < P}} d. \quad (4.8)$$

Proof. For $j \in F^+$, one has $P \nmid j$, $N_P(j) = 1$, and $N_n(j) = 0$. Hence $e_P(j) = -1$, and the $d = P$ term in (4.2) is $-P$. For a proper divisor with $\sigma_d = -1$, (4.6) gives $\sigma_d de_d \leq d\rho_d$; for $\sigma_d = +1$, it gives $\sigma_d de_d \leq dp$. This proves (4.7).

At an anchor the $d = P$ term vanishes, while $|e_d(h)| \leq p$ for every proper divisor. Summing proves (4.8). $\square$

5 Proof of Theorem 1.1

Proof of Theorem 1.1. For $j \in F^+$, Lemmas 3.1 and 4.2 give

$$\begin{aligned}
\frac{w_j}{pC} &= u(j) - u_{j_p} \varepsilon(j_{q_1}, \dots, j_{q_k}) \\
&\leq -P + \sum_{\substack{d|P, d < P \\ \sigma_d = -1}} d\rho_d + p \sum_{\substack{d|P, d < P \\ \sigma_d = +1}} d \\
&\quad + \frac{p}{q_1 - 1} \sum_{\substack{d|P \\ d < P}} d \\
&= -P + \mathcal{B}_k.
\end{aligned} \tag{5.1}$$

Condition (1.9) makes this strictly negative. Lemma 3.1 gives $w = 0$ on H , and (3.4) gives $w \in V_n^\perp$. Proposition 2.1 therefore forces every nonnegative Φ_n -multiple of degree at most $(p-1)P$ to be supported on H . Lemma 2.2 makes it a positive scalar multiple of $R_{n,p}$. Since $R_{n,p}$ has degree $(p-1)P$, the degree bound and uniqueness follow. $\square$

6 Infinite families outside the reciprocal condition

Proof of Theorem 1.2. Fix $k \geq 3$, $0 < \delta < 1/k$, and suppose (1.11) holds. For sufficiently large p , every $q_i < 2p$. If the complement of the prime support of $d \mid P$ consists of one prime q_i , then $d = P/q_i$, $\sigma_d = -1$, and

$$\rho_d = q_i - p. \tag{6.1}$$

Its normalized contribution is

$$\frac{d\rho_d}{P} = \frac{q_i - p}{q_i} \leq \delta. \tag{6.2}$$

The k such terms contribute at most $k\delta < 1$. If the complement has size $c \geq 2$, both possible upper-bound contributions are at most dp , and hence

$$\frac{\text{upper-bound contribution}}{P} \leq \frac{dp}{P} = \frac{p}{P/d} \leq \frac{1}{p^{c-1}}. \tag{6.3}$$

All complement sizes at least two therefore contribute at most

$$\sum_{c=2}^k \binom{k}{c} p^{1-c} \longrightarrow 0. \tag{6.4}$$

The normalized anchor term is

$$\frac{p}{q_1 - 1} \left(\prod_{i=1}^k \left(1 + \frac{1}{q_i} \right) - 1 \right) \longrightarrow 0. \tag{6.5}$$

Combining the three estimates gives the uniform bound, over all prime tuples satisfying (1.11),

$$\frac{\mathcal{B}_k}{P} \leq k\delta + \sum_{c=2}^k \binom{k}{c} p^{1-c} + \frac{p}{q_1 - 1} \left(\left(1 + \frac{1}{p}\right)^k - 1 \right) = k\delta + O_k(p^{-1}). \quad (6.6)$$

Since $k\delta < 1$, condition (1.9) eventually holds uniformly. Conversely,

$$p \sum_{i=1}^k \frac{1}{q_i} \geq \frac{k}{1 + \delta} > 2, \quad (6.7)$$

so Steinberger's condition (1.5) fails.

The prime number theorem implies that $(p, (1 + \delta)p]$ contains at least k primes for every sufficiently large prime p , producing infinitely many tuples. Also $p_{m+i}/p_m \rightarrow 1$ for fixed i , which proves the consecutive-prime statement. $\square$

7 Four-prime corollary and exact calibration

For $k = 3$, write $(q_1, q_2, q_3) = (q, r, s)$.

Corollary 7.1. *Let $n = pqrs$ be a product of distinct odd primes $p < q < r < s < 2p$. If*

$$\begin{aligned} &qr(s - p) + qs(r - p) + rs(q - p) + p(q + r + s) + p \\ &+ \frac{p}{q - 1}(qr + qs + rs + q + r + s + 1) < qrs, \end{aligned} \quad (7.1)$$

then the conclusion of Theorem 1.1 holds.

Proof. For $d = qr, qs, rs$, the residues ρ_d are respectively $s - p, r - p, q - p$. For $d = 1$, use $\rho_1 \leq p$. The other proper-divisor terms agree with the bounds in (7.1), so its left side majorizes $\mathcal{B}_3$. $\square$

The exact arithmetic checker gives the following consecutive-prime examples:

Total prime factors	Prime tuple	n	$\mathcal{B}_k/P$
4	23, 29, 31, 37	765049	31795/32116
5	37, 41, 43, 47, 53	162490421	165666531/175665320
6	59, 61, 67, 71, 73, 79	98733594781	96500955779/100407045540
7	97, 101, 103, 107, 109, 113, 127	168897325606883	149066984374577/174120954233900

Every row satisfies $\mathcal{B}_k/P < 1$ and fails (1.5). These examples calibrate the formulas and are not used in the infinite-family proof.

8 Scope and relation to earlier work

Steinberger [1] introduced the problem, the separation criterion, and the reciprocal condition (1.5). Theorem 1.1 uses the same separation framework but a different explicit certificate; Theorem 1.2 reaches infinitely many tuples outside (1.5).

The companion fixed-parameter paper [3] proves the conjectured minimum and uniqueness for

$$46189 = 11 \cdot 13 \cdot 17 \cdot 19 \quad \text{and} \quad 96577 = 13 \cdot 17 \cdot 19 \cdot 23$$

using exact stored integer certificates. Those small tuples need not satisfy (1.9), so the present theorem does not logically subsume the fixed-parameter results. The fixed certificates motivated the projection construction, whereas the present work extracts a symbolic mechanism for infinitely many additional cases.

Reddy [4] treats lowest-degree 0, 1-polynomials and minimal vanishing sums for even orders with at most three distinct prime factors. Kepka and Korbelař [5] treat bounds for nonnegative multiples of a general polynomial and determine the cubic case. Neither directly supplies (1.8), the anchor-corrected CRT projection, or Theorem 1.2.

This paper does not prove or disprove Steinberger’s conjecture in full. It does not claim that (1.9) is necessary or that the final-interval ansatz works for every prime tuple. Failure of (1.9) is only failure of this sufficient estimate, not evidence of a counterexample.

The novelty and open-status check was refreshed on 27 August 2026. We searched arXiv, OpenAlex, Semantic Scholar, and general scholarly web indexes using the title and DOI of [1], the exact minimum-degree problem, and combinations of “cyclotomic”, “nonnegative coefficients”, “tail projection”, “divisor residue”, and “separation certificate”. We also inspected the forward-citation neighborhood of [1] and the 2025 journal version of [2]. We found no source directly stating (1.8), the anchor-corrected final-interval projection, or Theorem 1.2. This is a bounded literature search, not a proof of global priority; differently phrased, unindexed, or concurrent work may still narrow the contribution.

9 Reproducibility

The proof is symbolic and does not depend on finite enumeration or floating-point optimization. The accompanying checker `verification/verify_general_tail_condition.py` uses exact integer and rational arithmetic to verify primality and factor ordering, evaluate every proper-divisor term in $\mathcal{B}_k$, test (1.9), and test Steinberger’s reciprocal condition. For example:

```
python -X utf8 '
verification\verify_general_tail_condition.py' '
--primes 37 41 43 47 53
```

The script rejects inputs outside the theorem’s domain, including even, composite, repeated, or unordered factors. It is a calibration aid, not a substitute for the proof.

10 AI-assisted research and writing disclosure

Computational exploration, proof organization, adversarial checking, and language editing were assisted by OpenAI Codex. Carptopus selected and reviewed the mathematical claims, verified the retained evidence, and assumes responsibility for the manuscript.

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